

API Idempotency and Safe Retries

Backend development study note for document retrieval and grounded question-answering practice.

Idempotency

An operation is idempotent when repeating the same request has the same intended effect as performing it once. HTTP defines GET, PUT, and DELETE as idempotent methods, although application behavior still matters.

Idempotency keys

For operations such as payment creation, an API can accept an idempotency key. The server records the result associated with the key so retries do not create duplicate effects.

Timeout ambiguity

A timeout does not prove that the server did not perform an operation. The server may have completed the operation while the response was delayed or lost. Clients should use idempotency mechanisms or check operation status before retrying non-idempotent actions.